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1 The Prime Structure Factor

1.0.1 The primes diffract like a crystal — and the crystal is the wheel

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What this is. A clean empirical/exact result (the diffraction pattern of the primes) with a clearly-flagged interpretive reading (the ED “interrupted wave = particle” picture). The object is standard — the prime *structure factor*, a.k.a. “primes as a quasicrystal” (Torquato–Zhang–de Courcy-Ireland, 2018); the value here is arriving at it from the particle/point-process side and reading its peaks directly as the wheel. Tags: **[fact]** established/exact · **[emp]** measured here · **[interp]** interpretation. Companion (position-space dual): `Offset_Correlation_Curve.md`.

1.1 1. The object

Treat the primes as a **point process** — a stream of discrete events on the line — and ask what pattern they would make in a diffraction experiment. That is the **structure factor**

$$S(k) = \frac{1}{M} \left| \sum_{p \leq N} e^{2\pi i k p} \right|^2,$$

the squared coherent amplitude at spatial frequency k (cycles per integer), over the M primes in the window. A featureless (Poisson) point process gives $S(k) \approx 1$; sharp peaks that **grow with** M signal genuine crystalline order at that frequency.

1.2 2. The result: Bragg peaks at every rational, with height $(\mu(m)/\varphi(m))^2$

Measured over the 148,932 odd primes below 2×10^6 (Figure [figures/prime_structure_factor.png](#)), the primes produce **sharp Bragg peaks at rational frequencies**, and the peak at $k = a/m$ (lowest terms) has height

$$\frac{S(a/m)}{M} = \left(\frac{\mu(m)}{\varphi(m)} \right)^2 \quad [\text{fact, exact}].$$

[emp] matching to the digit:

$k = a/m$	S/M	$(\mu(m)/\varphi(m))^2$	what it is
1/2	1.0000	$(-1/1)^2 = 1$	all-odd — <i>perfect</i> period-2 order (tallest possible)
1/3, 2/3, 1/6, 5/6	0.2500	$(\mp 1/2)^2 = 1/4$	the 2 · 3 wheel
1/5, ..., 1/10, ...	0.0625	$(1/4)^2 = 1/16$	the factor-5 structure
5/14, ...	0.0278	$(1/6)^2 = 1/36$	the factor-7 structure
1/4, 1/8, 3/8, ...	0.00000	$\mu(4) = \mu(8) = 0$	suppressed — powers of 2 add no order

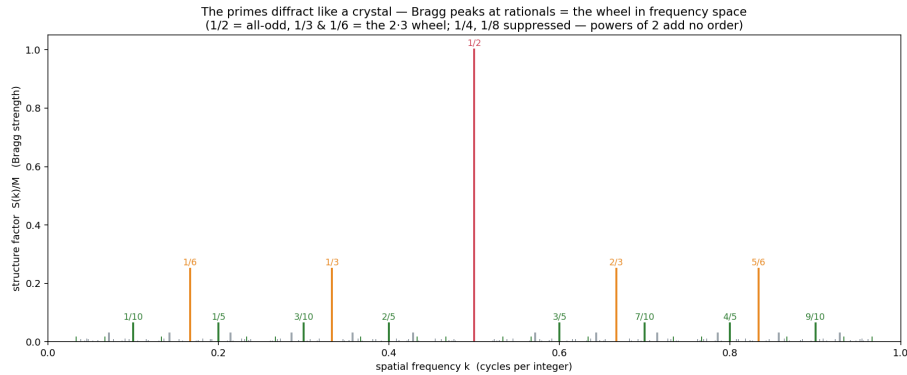


Figure 1: The prime structure factor: sharp Bragg peaks at rational frequencies — the wheel in frequency space. Tallest at 1/2 (all-odd), the 2 · 3 wheel at 1/3, 1/6, 2/3, 5/6, the factor-5 structure lower, and silence at the powers of 2. Heights follow $(\mu(m)/\varphi(m))^2$.

Two things make this remarkable:

- **The heights scale with M** (the $1/2$ peak is literally $S = M$), so these are *true Bragg peaks* — real crystalline order, not noise. The primes have **no period whatsoever, yet diffract like a crystal**: the definition of a **quasicrystal** (effectively limit-periodic).
- **The height law is the wheel — and *only* the wheel**. The height is $(\mu(m)/\varphi(m))^2 = \mu(m)^2/\varphi(m)^2$, so what survives the squaring is $\mu(m)^2 \in \{0, 1\}$ — a bare **squarefree indicator** (powers of 2, and 9, 25, ... give **no peak**) — times $1/\varphi(m)^2$. That is the coprimality/singular-series *template*, written in frequency space. **The *sign* of $\mu(m)$ — the parity / Möbius-randomness / Sarnak-hard layer — is squared away and does not appear in $S(k)$ at all**. So the structure factor probes exactly the **wheel/template** (the finite-memory-*reachable* layer) and is **blind to the parity barrier**: it sees the order the sieve builds, never the hard randomness it hides. (*Correction noted 2026-07-14, prompted by the ED-session diffraction analysis.*)

1.3 3. The Fourier dual of the offset comb

The offset-correlation comb and this structure factor are **Fourier transforms of each other** (Wiener–Khinchin): the diffraction pattern is the transform of the pair correlation. So the same wheel shows twice, once in each space:

	position space (the comb)	frequency space (this)
multiples of 6	tall teeth	peaks at $1/6, 1/3, 2/3, 5/6$
factor 5 (30)	tallest teeth	peaks at fifths/tenths
powers of 2	flat baseline	silence at $1/4, 1/8, \dots$

You can read the wheel off either side of the Fourier mirror.

1.4 4. The structure factor splits order from randomness

The diffraction pattern separates the two faces of the primes cleanly, in one image:

- the **sharp Bragg peaks = the ordered part = the wheel** — deterministic, crystalline, exact;
- the **diffuse background = the disordered part = the pseudo-randomness** — the zero-side, the Cramér noise.

Everything deterministic about the primes lives in the spikes; everything random lives in the haze. This is the sharpest statement of the collection’s thesis — *wheel order plus pseudo-random disorder* — now as a physical diffraction pattern.

1.5 5. The particle reading (ED)

[interp] The picture that led here: *an interrupted wave is a particle; the primes are the interruptions (the “did-happen” commitments); show me their particle spectrum.* This is that spectrum. Treated as a stream of particles, the primes are not scattered at random — they sit on a **hidden quasicrystalline lattice, and the lattice is the wheel**. Each prime is a commitment event; the events, collectively, diffract.

Honest boundary, as always: this is a **structural** result about a point process, not a claim that primes are physical particles. What is shared with the ED/particle picture is the *point-process form* — a continuous intensity (the wave, $1/\log x$) punctuated by discrete events (the particles/primes), whose collective arrangement has a diffraction pattern. That form is exact and measured; the physics vocabulary is a lens, not an identity.

1.6 6. The sixth shadow

The wheel, once more — now as a crystal. The shadows so far: jumping champions, the Seven Sisters, the forbidden widths $\{2, 4, 6, 10\}$, the angle-wobble 1%, the offset-correlation comb, and this **diffraction pattern**. Six faces, one wheel; this one is the primes revealed as a quasicrystal made of commitments.

1.7 Appendix. Reproduction

```
import numpy as np
def sieve(n):
    b=bytearray([1]*(n+1); b[0]=b[1]=0; i=2
    while i*i<=n:
        if b[i]: b[i*i::i]=bytearray(len(b[i*i::i]))
        i+=1
    return b
N=2_000_000; B=sieve(N)
P=np.array([i for i in range(3,N+1) if B[i]],dtype=float) # odd primes
M=len(P)
def SN(k): return abs(np.sum(np.exp(2j*np.pi*k*P)))*2/M/M # structure factor / M
from math import gcd
# Bragg peak at a/m has height (mu(m)/phi(m))^2 ; compare:
for m in (2,3,4,5,6,7,10):
    a=1
    print(m, round(SN(a/m),4)) # 1, 0.25(if m=3), 0(m=4), 0.0625(m=5), 0.25(m=6)...
```

1.8 References

- S. Torquato, G. Zhang, M. de Courcy-Ireland, *Uncovering multiscale order in the prime numbers via scattering* (J. Stat. Mech., 2018) — the prime

structure factor / “prime quasicrystal.”

- Wiener–Khinchin: the structure factor is the Fourier transform of the pair correlation.
- A. Proxmire, *The Offset-Correlation Curve* (this repository) — the position-space dual.
- A. Proxmire, *One Wheel, Many Shadows* and *The Wolves and the Clock* (Primes collection) — the unifying frame and the point-process reading.
- Figure: `figures/prime_structure_factor.png`.